

Review

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Classroom: Teaching Center 401

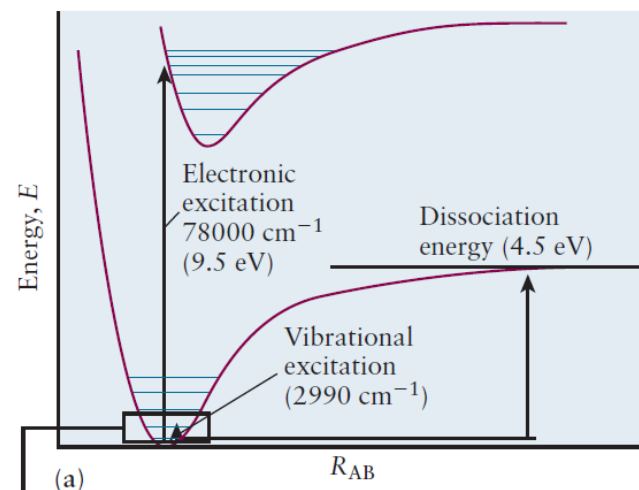
Wednesday/Friday: 10:15-11:00; 11:10-11:55

2026/01/04

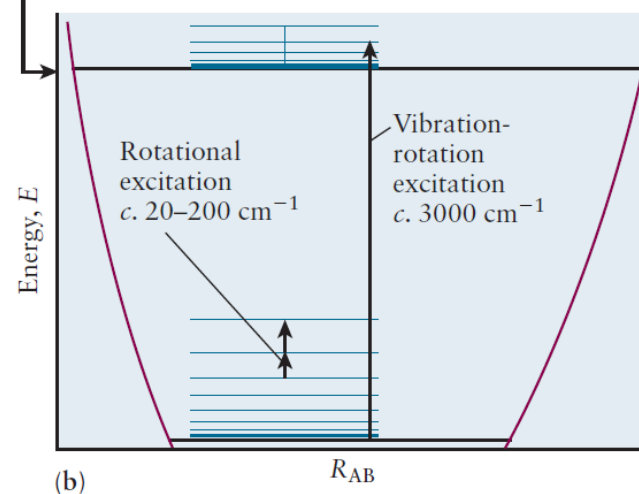
Introduction to Molecular Spectroscopy

Born–Oppenheimer approximation: slow nuclei, fast electrons.

$$E_{\text{total}} = E_{\text{trans}} + E_{\text{rot}} + E_{\text{vib}} + E_{\text{el}}$$



Absorption from the ground electronic state to an excited electronic state



Vibrational levels in the ground electronic state showing pure rotation, pure vibration, and vibration-rotation transitions

Intensities of Spectral Transitions

The intensity of the transmitted beam: **Beer–Lambert law**

$$I_t = I_0 e^{-\alpha \ell}$$

I_0 : intensity of the incident beam

I_t : intensity of the transmitted beam

ℓ : path length

α : absorption coefficient for the sample

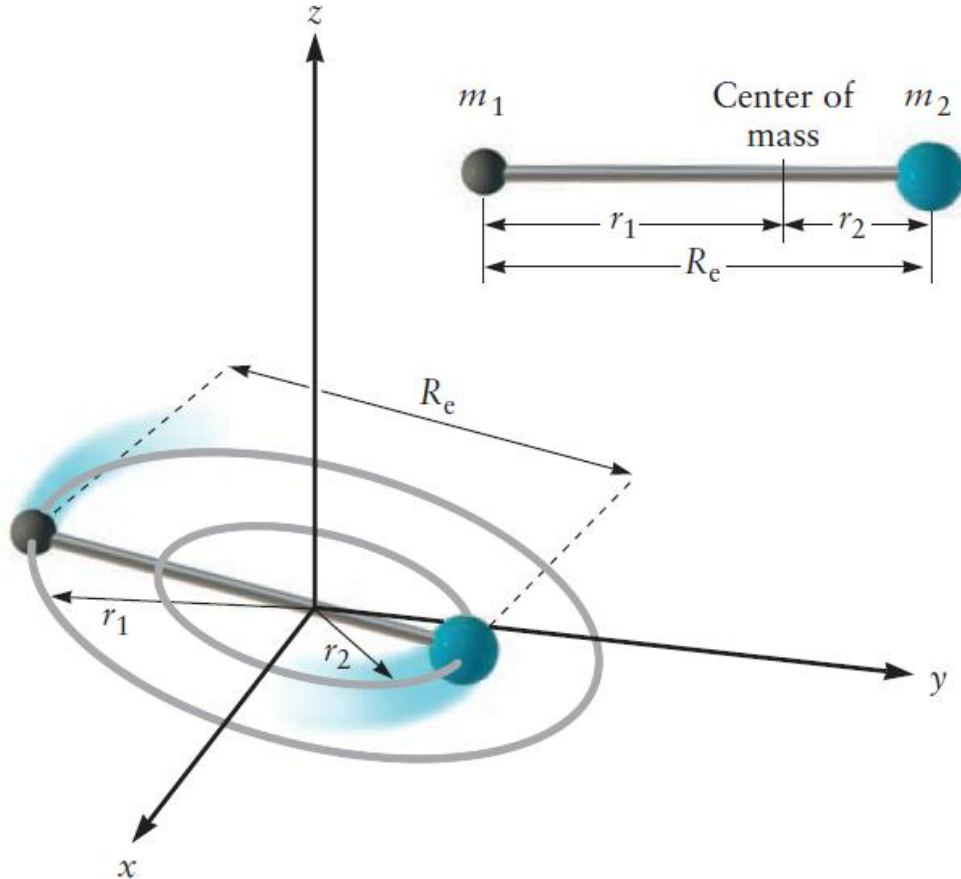
The Beer–Lambert law (first order kinetics): a constant fraction of the light is absorbed per unit path length and that fraction is independent of the intensity of the incident radiation

$$I_t = I_0 \exp[-(N/V)\sigma \ell]$$

N/V : the concentration of light-absorbing molecules

Diatomic Molecules: Rotational Spectroscopy

Diatomic molecules: **linear rigid rotor model 刚性转子** (the bond length is fixed; the molecule rotates about an axis that is perpendicular to the internuclear axis and passes through its center of mass)



The rotational motion is described by a **single moment of inertia 单惯性矩**

$$I = \mu R_e^2$$

R_e : equilibrium bond length

$\mu = m_1 m_2 / (m_1 + m_2)$, reduced mass

$$r_1 = \left(\frac{m_2}{m_1 + m_2} \right) R_e \text{ and } r_2 = \left(\frac{m_1}{m_1 + m_2} \right) R_e$$

Diatomic Molecules: Rotational Spectroscopy

The **wave functions for the rigid rotor** are identical to the angular parts of the hydrogen atom wave functions (the angular momentum is quantized).

The square of the angular momentum
angular momentum projection along the z-axis

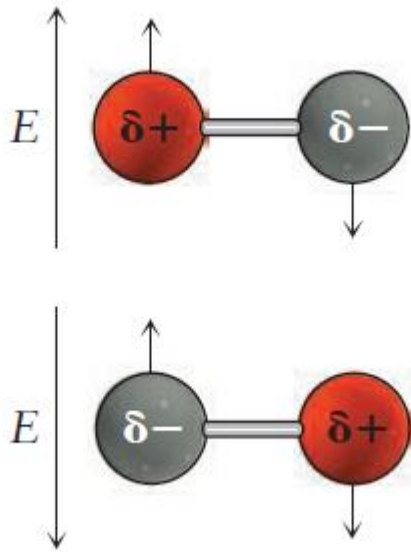
$$J^2 = J(J + 1)(h/2\pi)^2 \quad J = 0, 1, 2, 3...$$
$$J_z = M_J(h/2\pi) \quad M_J = -J, -J + 1, \dots, 0, \dots, J - 1, J$$

The energy levels for the linear rigid rotor

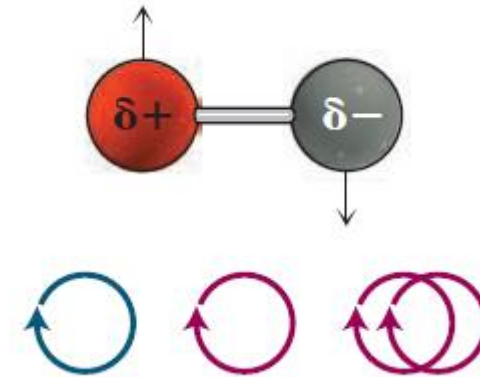
$$E_J = hc\tilde{B}J(J + 1) \quad \tilde{B} = (h/8\pi^2 cI)$$
$$E_J = hBJ(J + 1) \quad B = (h/8\pi^2 I)$$

Diatomic Molecules: Rotational Spectroscopy

Conservation of angular momentum: $\Delta J = \pm 1$ for microwave absorption and emission.



oscillating dipole moment of a heteronuclear diatomic molecule as it rotates



The angular momentum of a molecule decreases by **one unit** when a photon is emitted, the angular momentum being **carried by the photon**.

Diatomic Molecules: Rotational Spectroscopy

Conservation of energy for microwave absorption and emission.

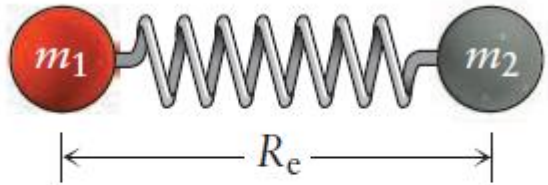
selection rule: the frequencies of the allowed transitions in wave numbers

$$\begin{aligned}\tilde{\nu} &= \tilde{B}[J_f(J_f + 1) - J_i(J_i + 1)] \\ &= \tilde{B}[(J_i + 1)(J_i + 2) - J_i(J_i + 1)] \\ &= 2\tilde{B}(J + 1)\end{aligned}$$

J_i and J_f represent the rotational quantum numbers of the initial and final states

Diatomic Molecules: Vibrational Spectroscopy

two masses connected by a spring



Diatomic molecule represented by a ball and spring model.

reduced mass of the system $\mu = m_1 m_2 / (m_1 + m_2)$

Hooke's law $F(R - R_e) = -k(R - R_e)$

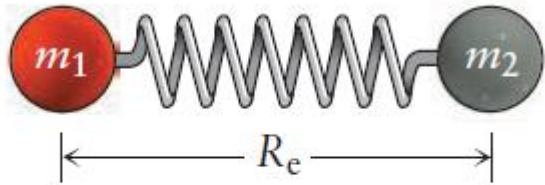
k: force constant (measure the stiffness of the bond)

$$V(R - R_e) = \frac{1}{2} k(R - R_e)^2$$

Harmonic oscillator approximation: a characteristic frequency $\nu = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}}$

Diatomic Molecules: Vibrational Spectroscopy

two masses connected by a spring



Diatomic molecule represented by a ball and spring model.

one-dimensional Schrödinger equation with the harmonic oscillator potential energy term added to the kinetic energy term.

$$-\frac{h^2}{8\pi^2m} \frac{d^2\psi(x)}{dx^2} + \frac{1}{2}kx^2\psi(x) = E\psi(x)$$

The quantized energy levels of the harmonic oscillator

$$E_v = \left(v + \frac{1}{2}\right) h\nu$$

ν is its fundamental vibrational frequency and v is a quantum number **0, 1, 2, ...**

Diatomic Molecules: Vibrational Spectroscopy

E does not equal to 0 even at 0 K;
this motion is called **zero-point motion**;
the energy is called the **zero-point energy**.

TABLE 20.2

The First Four Harmonic Oscillator Wave Functions

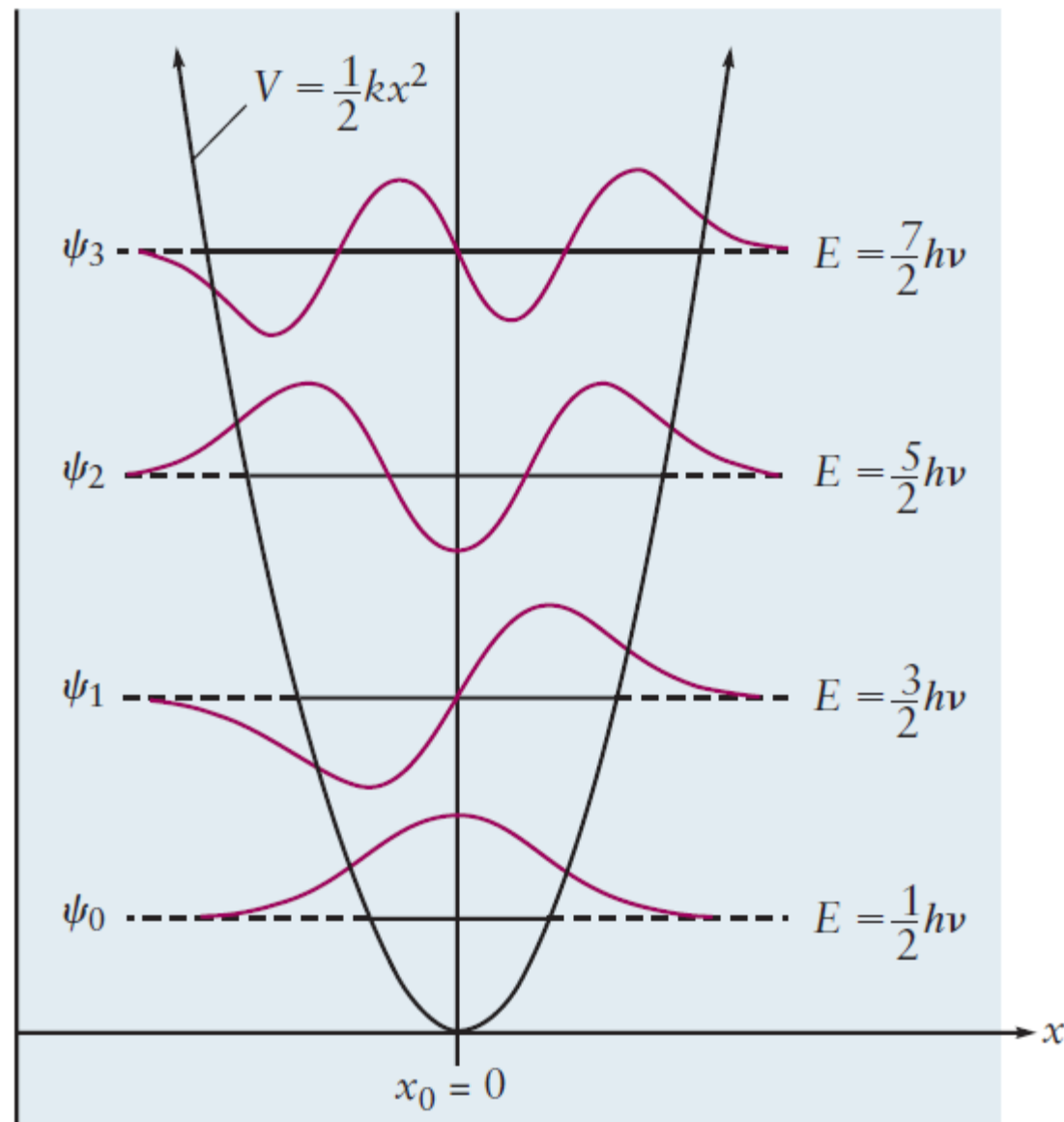
$$\alpha = \frac{2\pi}{h} (k\mu)^{1/2}$$

$$\psi_0(x) = \left(\frac{\alpha}{\pi}\right)^{1/4} e^{-\alpha x^2/2}$$

$$\psi_1(x) = \left(\frac{4\alpha^3}{\pi}\right)^{1/4} x e^{-\alpha x^2/2}$$

$$\psi_2(x) = \left(\frac{\alpha}{4\pi}\right)^{1/4} (2\alpha x^2 - 1) e^{-\alpha x^2/2}$$

$$\psi_3(x) = \left(\frac{\alpha^3}{9\pi}\right)^{1/4} (2\alpha x^3 - 3x) e^{-\alpha x^2/2}$$



Diatomic Molecules: Vibrational Spectroscopy

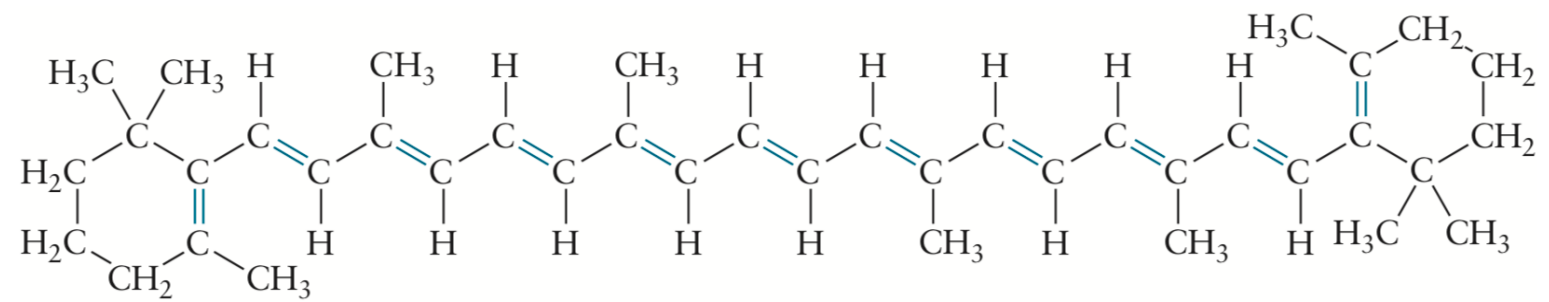
Energy differences between ground ($v = 0$) and first excited ($v = 1$) vibrational states: 2 kJ mol^{-1} to 40 kJ mol^{-1} ($>$ rotational energy differences; $<$ electronic energy differences)

1. These energy differences correspond to the absorption in the **infrared region** of the electromagnetic spectrum.
2. Strong absorption is observed only for transitions between states that differ by 1 in vibrational quantum number ($\Delta v = 1$).
3. Separation between adjacent energy levels is independent of quantum number: a single vibrational frequency to determine the force constant of the bond.

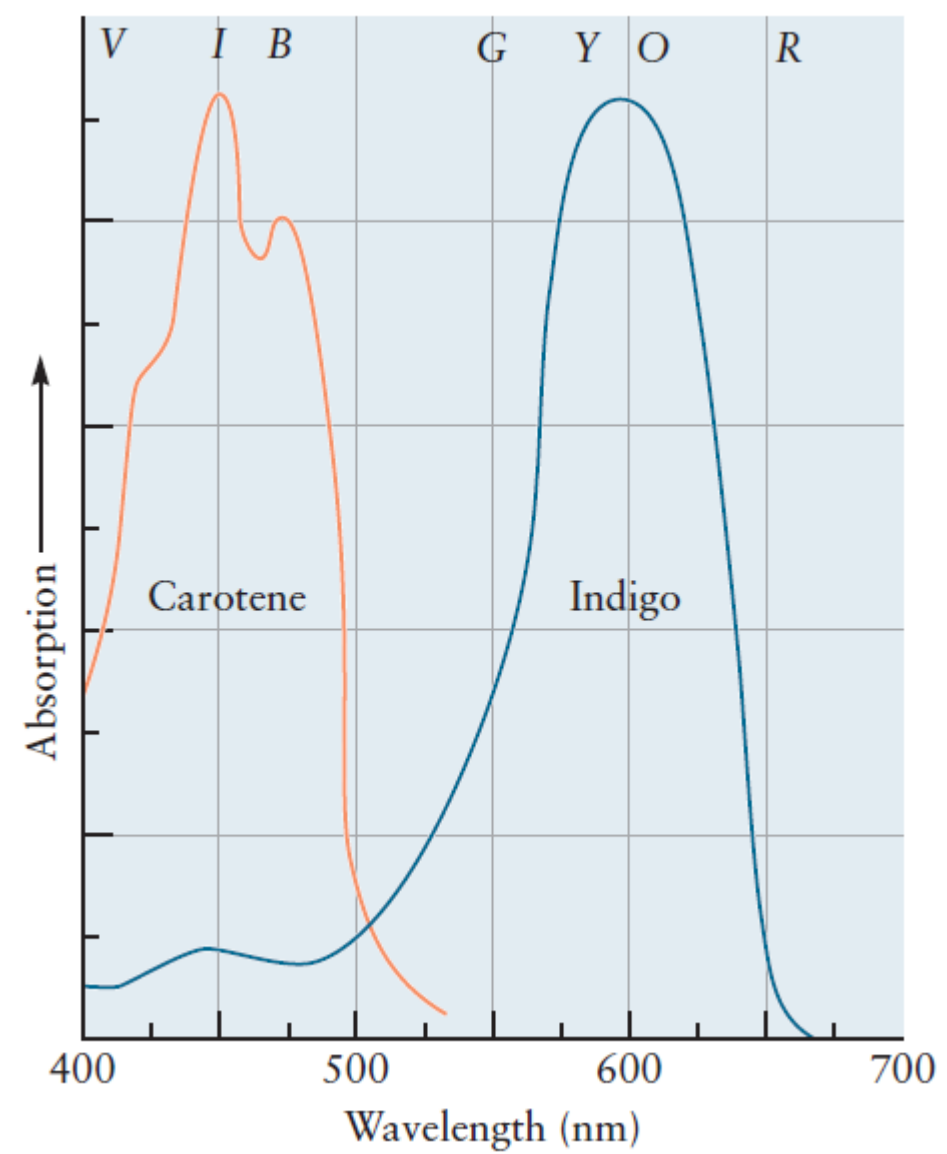
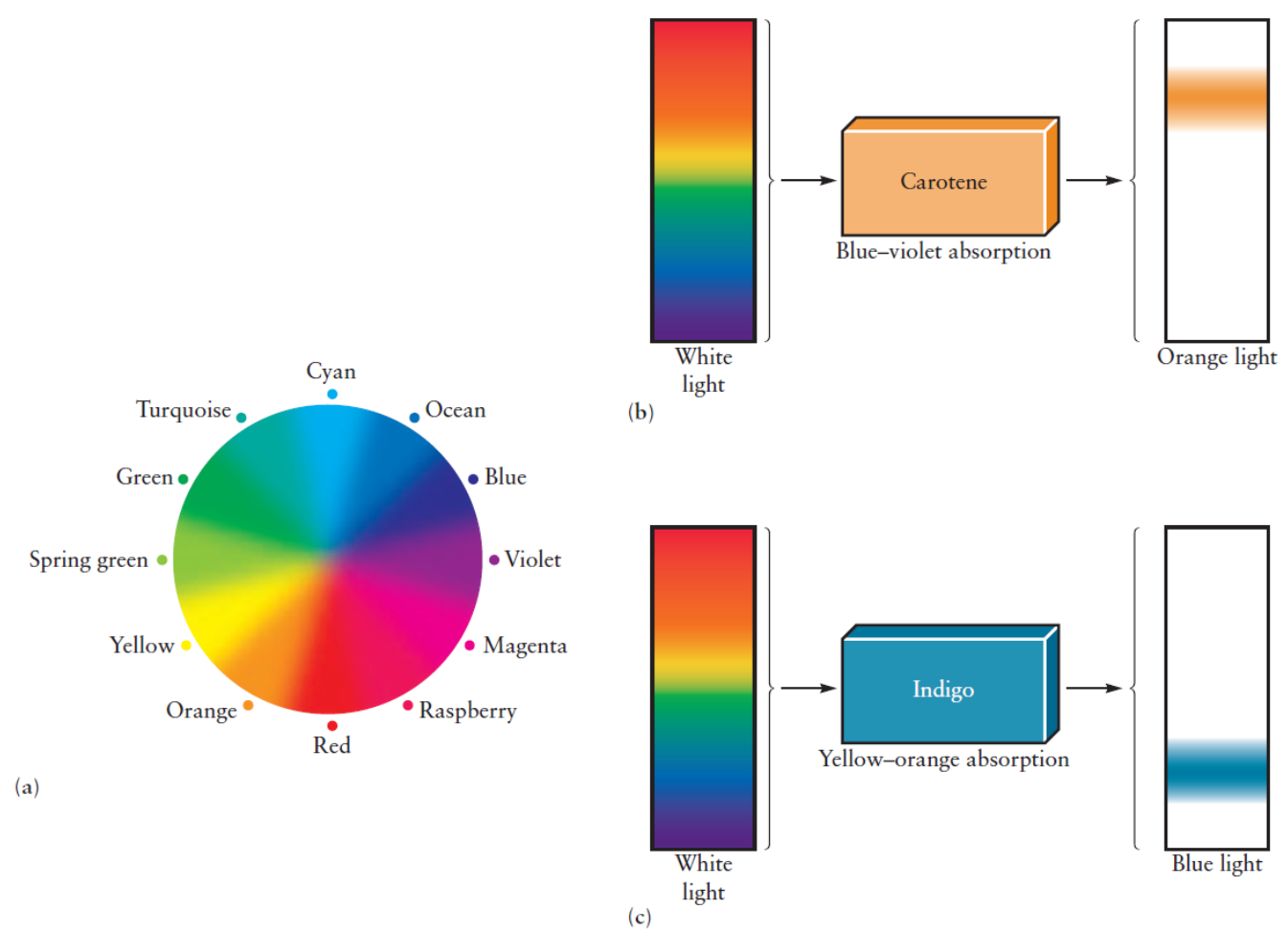
Electronic Spectroscopy

Now consider a molecule in which several double bonds alternate with single bonds in a conjugated π -bonding system like that for butadiene

T A B L E 20.2 Absorption of Light by Molecules with Conjugated π Electron Systems		
Molecule	Number of C=C Bonds	Wavelength of Maximum Absorption (nm)
C ₂ H ₄	1	162
C ₄ H ₆	2	217
C ₆ H ₈	3	251
C ₈ H ₁₀	4	304

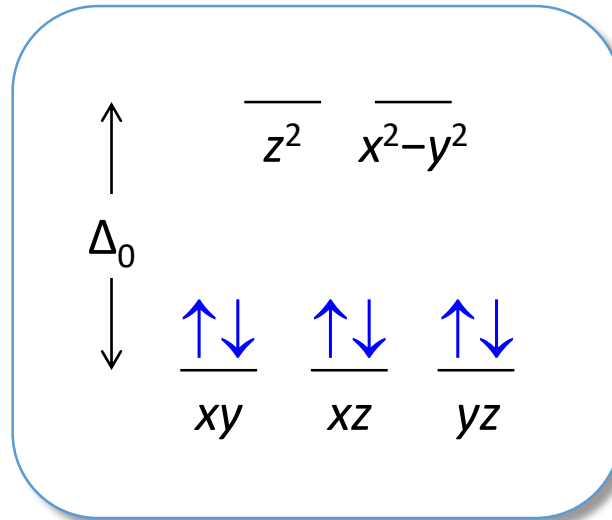
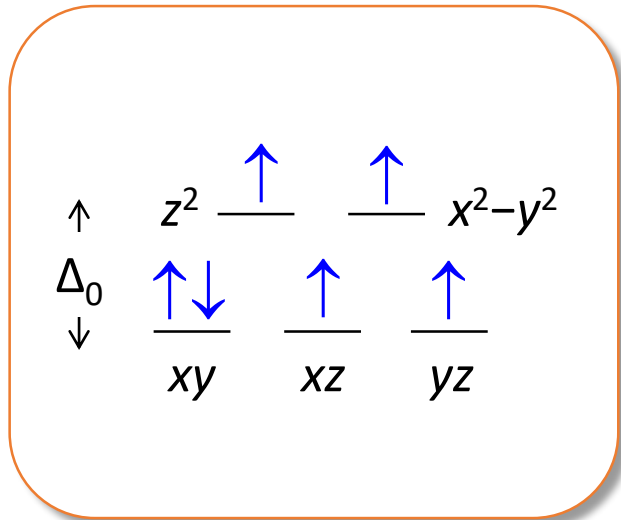


Electronic Spectroscopy



Optical Properties and the Spectrochemical Series

The wavelength of the strongest absorption is called λ_{max} and it is related to Δ_o

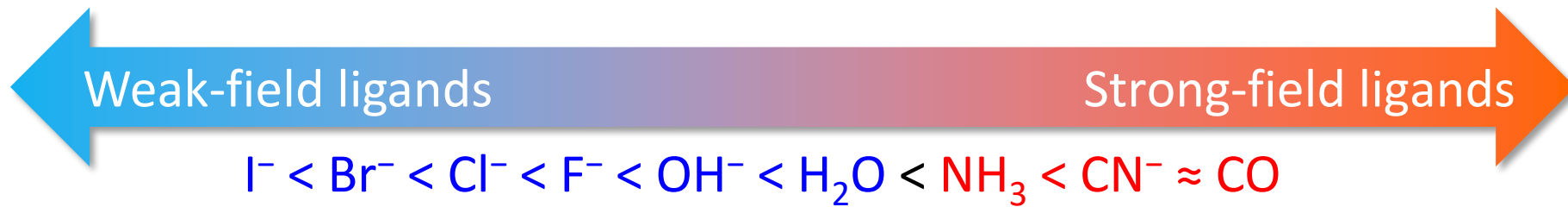


$$\Delta_o = h\nu = hc/\lambda_{\text{max}}$$

$[\text{Co}(\text{NH}_3)_6]^{3+}$	Orange
$[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$	A green form and a violet form
$[\text{Co}(\text{NH}_3)_5(\text{H}_2\text{O})]^{3+}$	Purple

High spin

Low spin



The smaller halide ions can approach the metal ion more closely: greater electrostatic repulsion.

Final

课程序号	课程名称	上课人数	考试日期	考试时间	考试地点	主考教师
CHEM1103.01	普通化学 I	65	2026-01-09	10:30-12:30	教学中心 202	米启兮
CHEM1103.02	普通化学 I	45	2026-01-09	10:30-12:30	教学中心 203	宁志军
CHEM1103.03	普通化学 I	65	2026-01-09	10:30-12:30	教学中心 204	谢珽
CHEM1103.04	普通化学 I	65	2026-01-09	10:30-12:30	教学中心 302	杨帆
CHEM1103.05	普通化学 I	65	2026-01-09	10:30-12:30	教学中心 303	刘朋昕

- **When: 2026/01/09 (Friday), 120 minutes (10:30-12:30), week 17**

- **Where: Teaching Center 204**

考试注意事项:

1. 随身携带学生证/身份证
2. 准备计算器!!!
3. 提供参考纸

Periodic Table of the Elements												1 H 1.00794		2 He 4.002602																													
												5 B 10.811		6 C 12.0107		7 N 14.00674		8 O 15.9994		9 F 18.9984032		10 Ne 20.1797																					
												13 Al 26.981538		14 Si 28.0855		15 P 30.973761		16 S 32.066		17 Cl 35.4527		18 Ar 39.948																					
3 Li 6.941		4 Be 9.012182		11 Na 22.989770		12 Mg 24.3050		19 K 39.0983		20 Ca 40.078		21 Sc 44.955910		22 Ti 47.867		23 V 50.9415		24 Cr 51.9961		25 Mn 54.938049		26 Fe 55.845		27 Co 58.933200		28 Ni 58.6934		29 Cu 63.546		30 Zn 65.39		31 Ga 69.723		32 Ge 72.61		33 As 74.92160		34 Se 78.96		35 Br 79.904		36 Kr 83.80	
37 Rb 85.4678		38 Sr 87.62		39 Y 88.90585		40 Zr 91.224		41 Nb 92.90638		42 Mo 95.94		43 Tc (98)		44 Ru 101.07		45 Rh 102.90550		46 Pd 106.42		47 Ag 107.8682		48 Cd 112.411		49 In 114.818		50 Sn 118.710		51 Sb 121.760		52 Te 127.60		53 I 126.90447		54 Xe 131.29									
55 Cs 132.90545		56 Ba 137.327		57 La 138.9055		72 Hf 178.49		73 Ta 180.9479		74 W 183.84		75 Re 186.207		76 Os 190.23		77 Ir 192.217		78 Pt 195.078		79 Au 196.96655		80 Hg 200.59		81 Tl 204.3833		82 Pb 207.2		83 Bi 208.98038		84 Po (209)		85 At (210)		86 Rn (222)									
87 Fr (223)		88 Ra (226)		89 Ac (227)		104 Rf (261)		105 Db (262)		106 Sg (263)		107 Bh (262)		108 Hs (265)		109 Mt (266)																											

Speed of light

$$c = 299\,792\,458 \text{ m}\cdot\text{s}^{-1}$$

Planck constant

$$h = 6.626\,07 \times 10^{-34} \text{ J}\cdot\text{s}$$

Vacuum permittivity

$$\epsilon_0 = 8.854\,19 \times 10^{-12} \text{ F}\cdot\text{m}^{-1}$$

Elementary charge

$$e = 1.602\,18 \times 10^{-19} \text{ C}$$

Electron mass

$$m_e = 9.109\,38 \times 10^{-31} \text{ kg}$$

Avogadro constant

$$N_A = 6.022\,14 \times 10^{23} \text{ mol}^{-1}$$

Bohr radius

$$a_0 = 0.529\,177 \times 10^{-10} \text{ m}$$

Rydberg constant

$$R_y = 13.605\,7 \text{ eV}$$

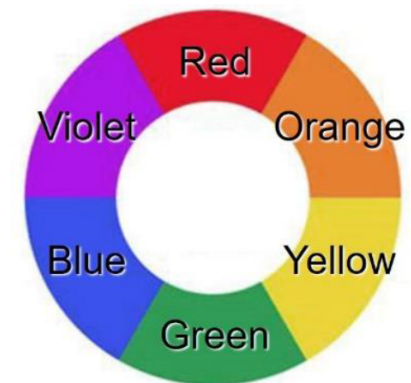
$$r_n = \frac{n^2}{Z} a_0$$

$$E_n = -\frac{Z^2}{n^2} R_y$$

$$\nu = \frac{1}{2\pi} \sqrt{k/m}$$

Visible spectrum

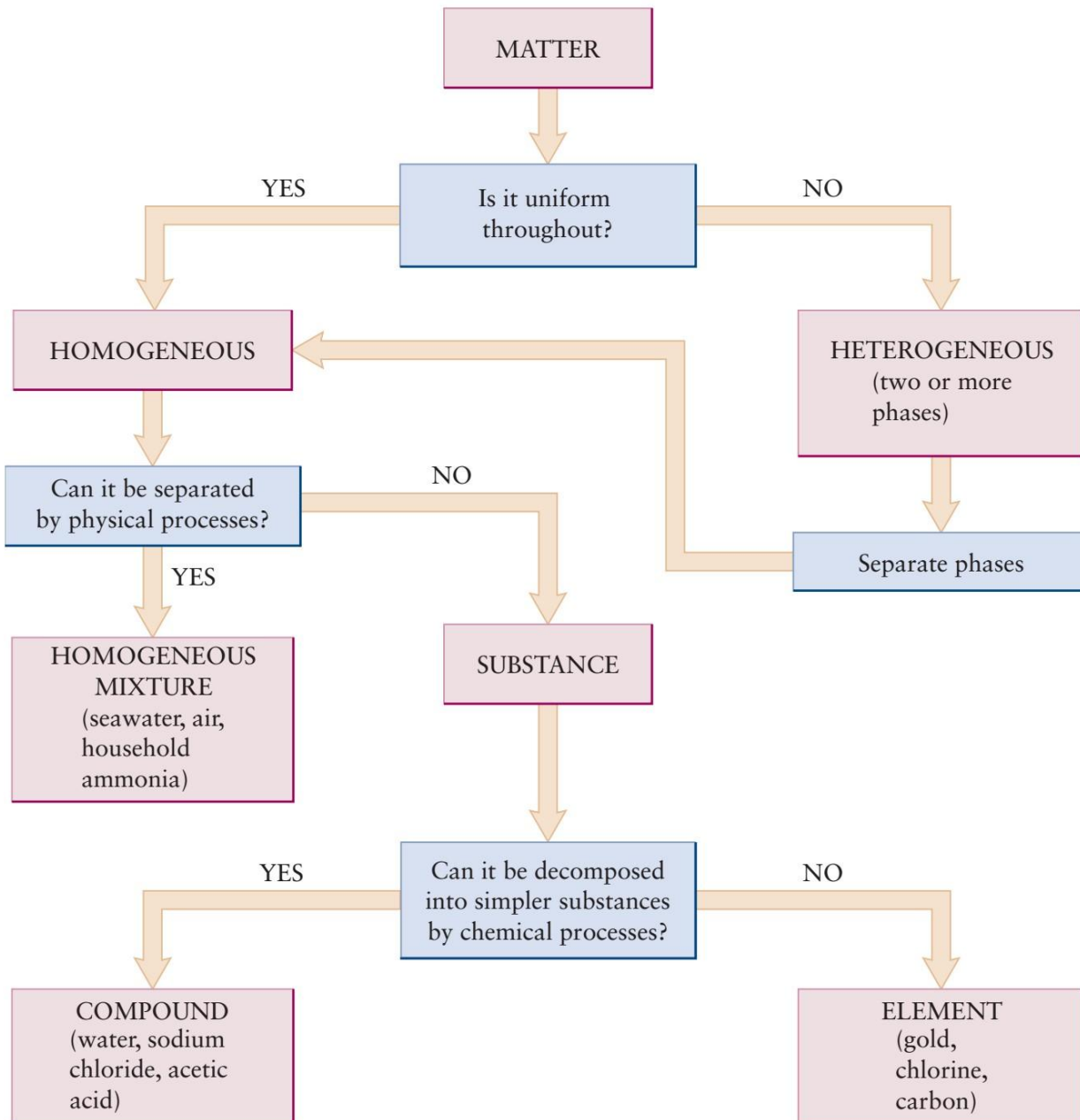
380	V	450	B	495	G	570	Y	590	O	620	R	750
380–450 nm	Violet	450–495 nm	Blue	495–570 nm	Green	570–590 nm	Yellow	590–620 nm	Orange	620–750 nm	Red	



Take-Home Message

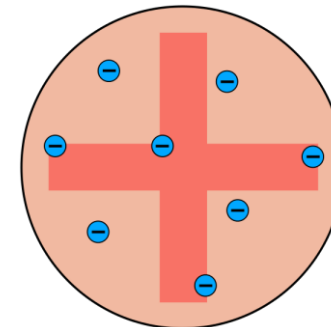
1. The atom in modern chemistry (chapter 1)
2. Chemical formulas, equations, and reaction yields (chapter 2)
3. Chemical bonding: the classical description (chapter 3)
4. Introduction to quantum mechanics (chapter 4)
5. Quantum mechanics and atomic structure (chapter 5)
6. Quantum mechanics and molecular structure (chapter 6)
7. Bonding in organic molecules (chapter 7)
8. Bonding in transition metal compounds and coordination complexes (chapter 8)
9. Molecular spectroscopy and photochemistry (chapter 20)

1. The atom in modern chemistry

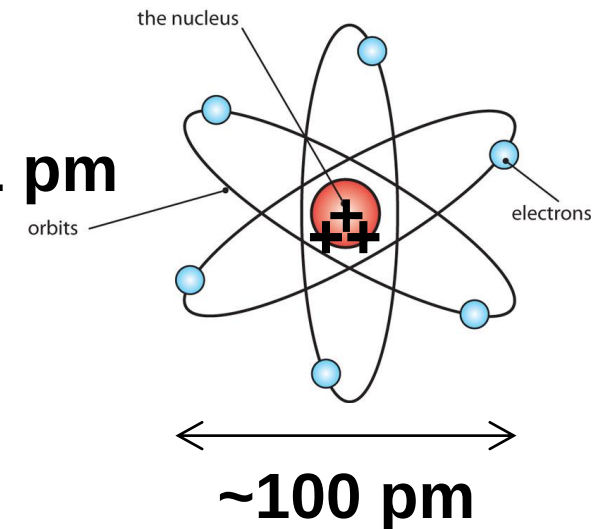


- **Laws of chemical combinations**

1. Law of Conservation of Mass
2. Law of Definite Proportions
3. Law of Multiple Proportions
4. Law of Combining Volumes
5. Avogadro's Law



< 0.01 pm



2. Chemical formulas, equations, and reaction yields

- The molecular formula is some whole-number multiple of the empirical formula.
- From Avogadro's hypothesis (same number of molecules of two gases in same volume, temperature and pressure), the ratio of molar masses of two gaseous compounds is the same as the ratio of their densities.

乙炔的摩尔质量

乙炔的密度

氧气的摩尔质量

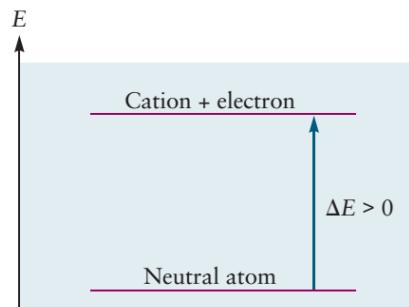
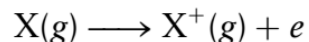
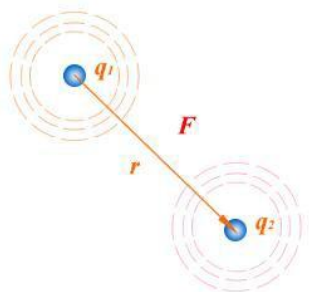
$$\text{Molar mass of welding gas} = \frac{1.06 \text{ g L}^{-1}}{1.31 \text{ g L}^{-1}} (32.0 \text{ g mol}^{-1}) = 25.9 \text{ g mol}^{-1}$$

氧气的密度

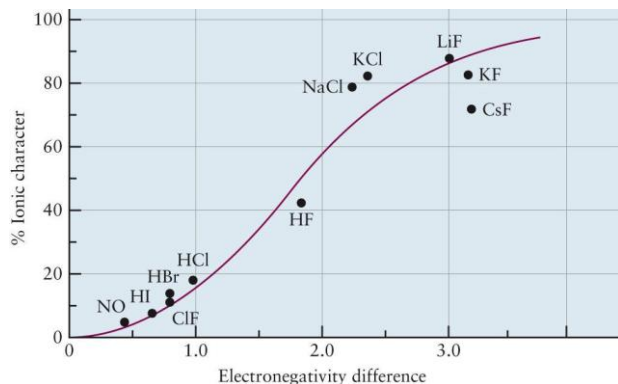
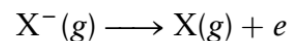
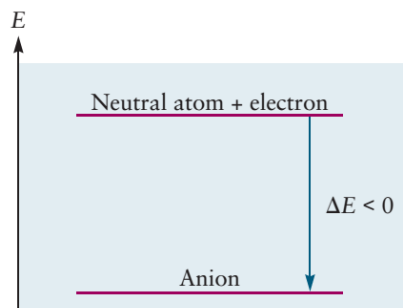
乙炔= C_2H_2

3. Chemical bonding: the classical description

Coulomb's law (库仑定律)



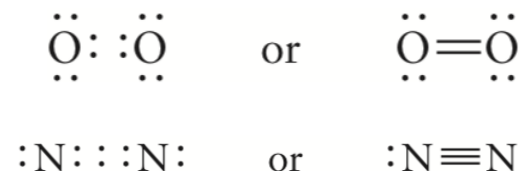
$$\Delta E = IE_1$$



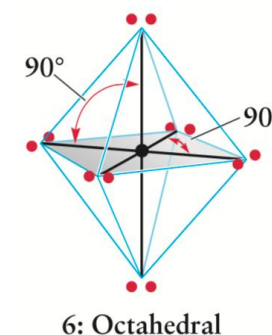
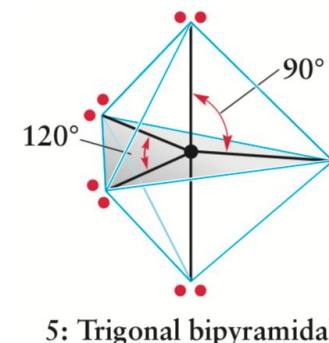
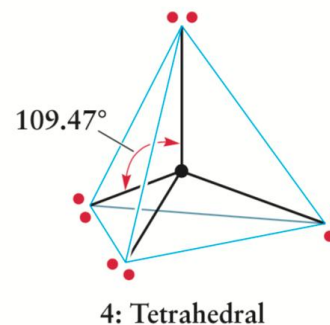
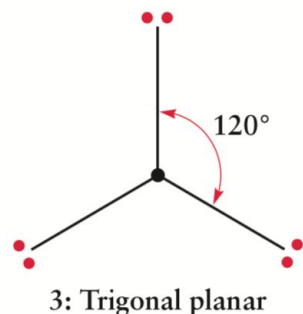
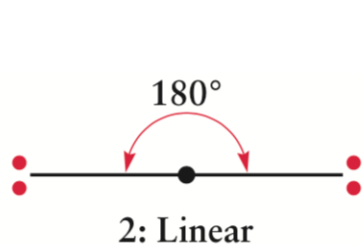
Ionic	Polar Covalent	Covalent
$\Delta\chi > 2.0$		$\Delta\chi < 0.4$

Lewis diagrams

- A **shared pair electrons** = a **covalent bond**
- A **lone pair electrons** = **no bonding**
- Each atom achieves its own **noble-gas shell** of electrons
- more than one pair may be shared (**double bond**, **triple bond**)

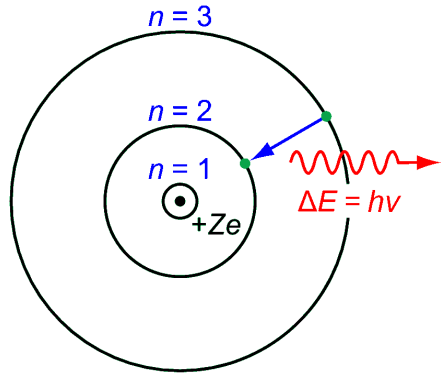


- **The Valence Shell Electron-Pair Repulsion Theory (VSEPR): 价层电子对互斥模型** $SN = \left(\text{number of atoms bonded to central atom} \right) + \left(\text{number of lone pairs on central atom} \right)$



4. Introduction to quantum mechanics

Bohr Model



$$L = m_e v r = n \frac{h}{2\pi} \quad n = 1, 2, 3, \dots$$

$$E_n = \frac{-Z^2 e^4 m_e}{8\epsilon_0^2 n^2 h^2} = -(2.18 \times 10^{-18} \text{ J}) \frac{Z^2}{n^2} \quad n = 1, 2, 3, \dots$$

the Schrödinger equation

$$\hat{H}\Psi = i\hbar \frac{\partial}{\partial t} \Psi$$

the Schrödinger equation (time-independent) $\hat{H}\psi = E\psi$

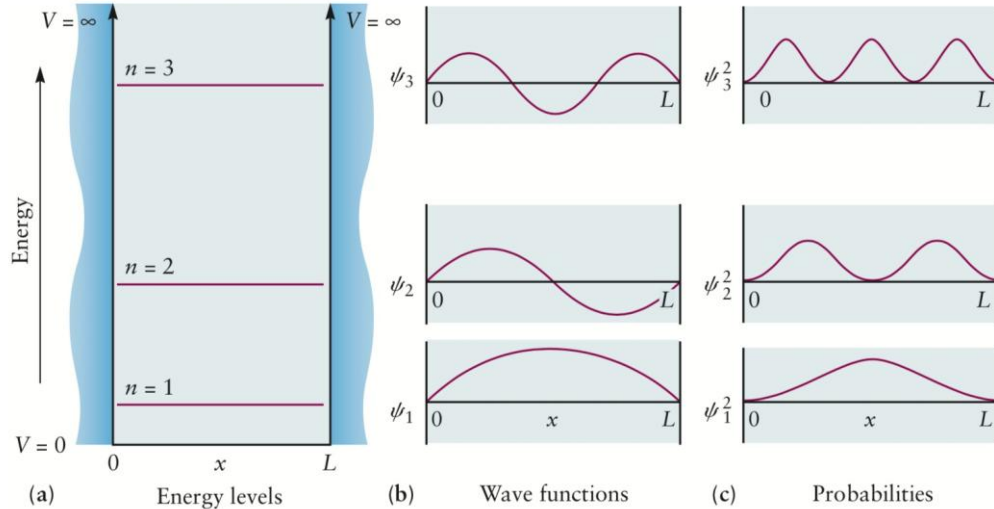
$$\hat{H} = \hat{T} + \hat{V} = \frac{\hat{p}^2}{2m} + V = -\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{r}, t). \quad \downarrow \quad \sum_{i=1}^n \frac{\partial^2}{\partial x_i^2}$$

$$-\frac{\hbar^2}{8\pi^2 m} \frac{d^2 \psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

the Schrödinger equation (time-independent, 1D)

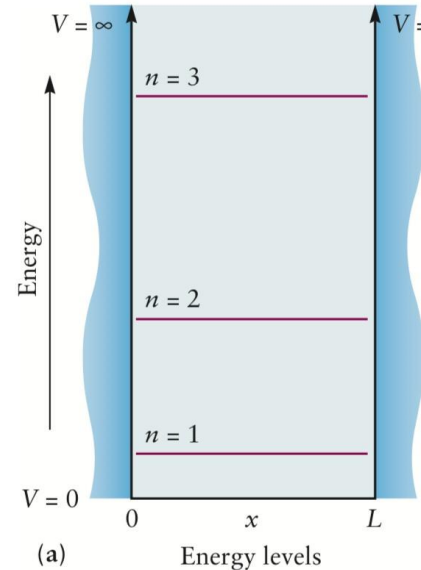
Probability of finding a particle is given by the square of the amplitude of the wave function

Probability: $P(x) = |\psi(x)|^2$



The normalized wave function for the one-dimensional particle in a box is

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \quad n = 1, 2, 3, \dots$$



$$E_n = \frac{n^2 h^2}{8mL^2} \quad n = 1, 2, 3, \dots$$

5. Quantum mechanics and atomic structure

$$n = 1, 2, 3, \dots$$

主量子数

$$\ell = 0, 1, \dots, n - 1$$

角量子数

$$m = -\ell, -\ell + 1, \dots, 0, \dots, \ell - 1, \ell$$

磁量子数

$$m_s = +1/2 \text{ or } -1/2$$

自旋量子数

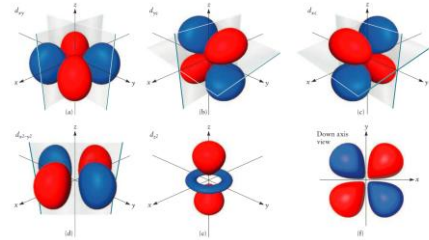
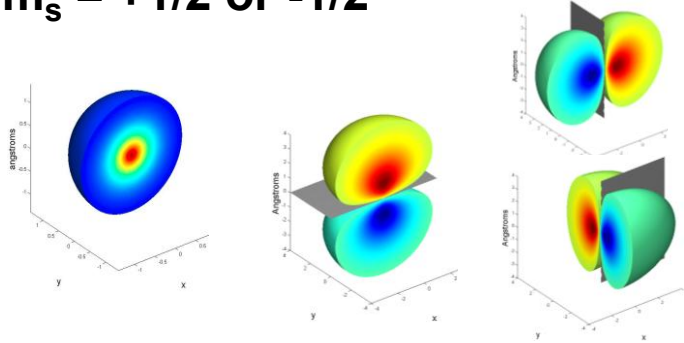
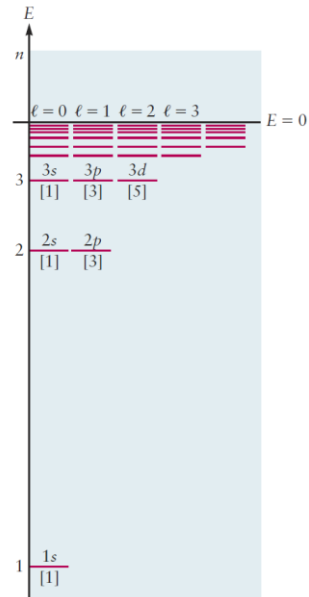


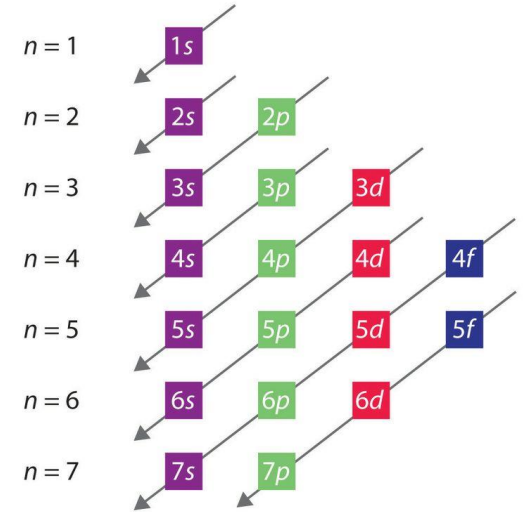
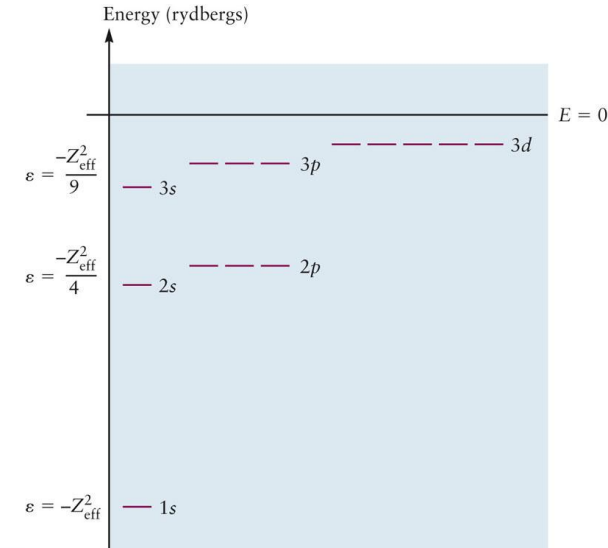
FIGURE 5.13 The shapes of the five 3d orbitals, with phases and nodal surfaces indicated. The "down axis" view shows the shapes of the first four orbitals (a)–(d) when viewed down the appropriate axis; the specific example shown here is the d_{xy} orbital viewed down the z -axis.

TABLE 5.2 Angular and Radial Parts of Wave Functions for One-Electron Atoms

Angular Part $Y(\theta, \phi)$	Radial Part $R_{nl}(r)$
$\ell = 0 \left\{ Y_0 = \left(\frac{1}{4\pi} \right)^{1/2} \right.$	$R_{1s} = 2 \left(\frac{Z}{a_0} \right)^{3/2} \exp(-\sigma)$
$\ell = 1 \left\{ \begin{aligned} Y_{10} &= \left(\frac{3}{4\pi} \right)^{1/2} \sin \theta \cos \phi \\ Y_{11} &= \left(\frac{3}{4\pi} \right)^{1/2} \sin \theta \sin \phi \\ Y_{12} &= \left(\frac{3}{4\pi} \right)^{1/2} \cos \theta \end{aligned} \right.$	$R_{2s} = \frac{1}{2\sqrt{2}} \left(\frac{Z}{a_0} \right)^{3/2} (2 - \sigma) \exp(-\sigma/2)$
$\ell = 2 \left\{ \begin{aligned} Y_{20} &= \left(\frac{5}{16\pi} \right)^{1/2} (3 \cos^2 \theta - 1) \\ Y_{21} &= \left(\frac{15}{4\pi} \right)^{1/2} \sin \theta \cos \theta \cos \phi \\ Y_{22} &= \left(\frac{15}{4\pi} \right)^{1/2} \sin^2 \theta \sin 2\phi \\ Y_{23} &= \left(\frac{15}{16\pi} \right)^{1/2} \sin^2 \theta \cos 2\phi \end{aligned} \right.$	$R_{3s} = \frac{2}{81\sqrt{3}} \left(\frac{Z}{a_0} \right)^{3/2} (27 - 18\sigma + 2\sigma^2) \exp(-\sigma/3)$
	$R_{3p} = \frac{4}{81\sqrt{6}} \left(\frac{Z}{a_0} \right)^{3/2} (6\sigma - \sigma^2) \exp(-\sigma/3)$
	$R_{3d} = \frac{4}{81\sqrt{30}} \left(\frac{Z}{a_0} \right)^{3/2} \sigma^2 \exp(-\sigma/3)$
$\sigma = \frac{Zr}{a_0}$	$a_0 = \frac{e_0 \hbar^2}{\pi e^2 m_e} = 0.529 \times 10^{-10} \text{ m}$



Hartree's method: Each electron moves in **an effective field** created by the nucleus and all the other electrons



The Pauli exclusion principle: no two electrons in an atom can have the same set of four quantum numbers (n, ℓ, m, m_s).

Hund's rules: when electrons are added to Hartree orbitals of equal energy, **a single electron enters each orbital** before a second one enters any orbital. In addition, the lowest energy configuration is the one **with parallel spins**.

6. Quantum mechanics and molecular structure

Linear Combination of Atomic Orbitals (LCAO, 原子轨道线性组合):

selecting sums and differences (linear combinations) of atomic orbitals to generate the best **approximation** to each type of **molecular orbital**.

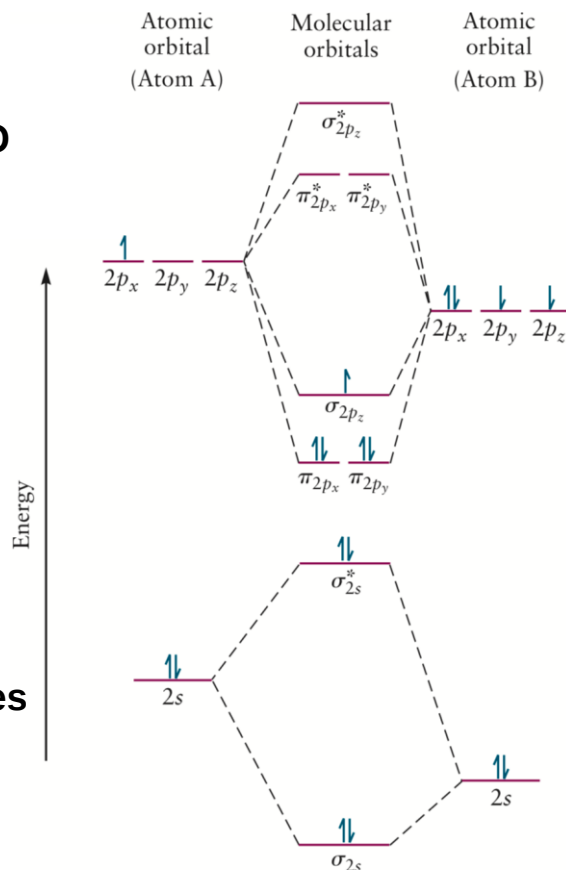
$$\psi_{\text{MO}}^{\text{cl}} = \sigma_{g1s}(1)\sigma_{g1s}(2) = [1s^{\text{A}}(1) + 1s^{\text{B}}(1)][1s^{\text{A}}(2) + 1s^{\text{B}}(2)]$$

Correlation diagram (轨道相关图): the energy-level diagram within the LCAO approximation.

- Obey the Aufbau Principle

Rules of bonding

- Maximum orbital overlap
- Compatible orbital symmetries
- Approximate atomic orbital energies



Valence bond (VB) model:

- Grew out of the **qualitative Lewis electron-pair model** in which the chemical bond is described as **a pair of electrons that is localized between two atoms**.
- The VB model **constructs a wave function for the bond** by assuming that **each atom arrives with at least one unpaired electron in an AO**.

$$\psi_{\text{VB}}^{\text{cl}} = 1s^{\text{A}}(1)1s^{\text{B}}(2) + 1s^{\text{A}}(2)1s^{\text{B}}(1)$$

Promotion (激发)

Hybridization (杂化)

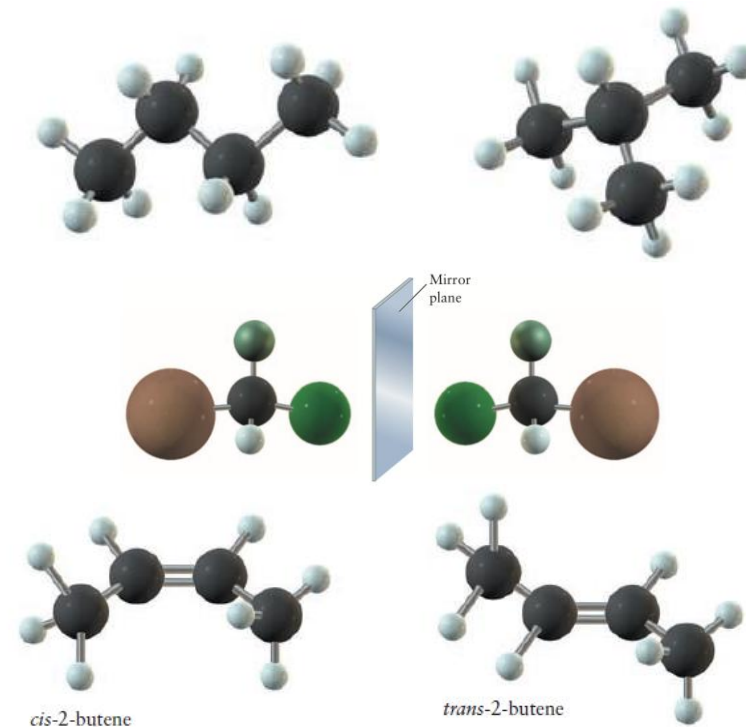
TABLE 6.5

Molecular Shapes Predicted by the Valence Shell Electron-Pair Repulsion Theory and Rationalized by Orbital Hybridization

Molecule	Steric Number	Number of Lone Pairs	Orbital Hybridization	Predicted Geometry	Image	Example
AX ₂	2	0	sp	Linear		BeH ₂ , CO ₂
AX ₃	3	0	sp ²	Trigonal planar		BF ₃ , SO ₃
AX ₂	3	1	sp ²	Bent		SO ₂
AX ₄	4	0	sp ³	Tetrahedral		CF ₄ , SO ₄ ²⁻
AX ₃	4	1	sp ³	Trigonal pyramidal		NH ₃ , PF ₃ , AsCl ₃
AX ₂	4	2	sp ³	Bent		H ₂ O, H ₂ S, SF ₂

7. Bonding in organic molecules

- **Geometrical isomers 几何异构体/同分异构体**: same molecular formula but a **different bonding structure**
- **Optical isomerism 光学异构体**: the two forms rotate the plane of polarized light in different directions
- **顺反异构体**: **cis (顺式)** and **trans (反式)** forms



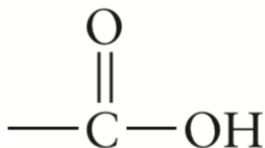
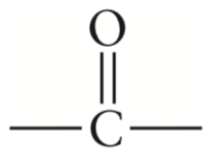
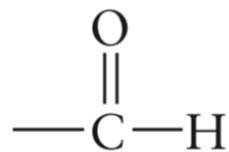
Alkanes (烷烃) vs. Alkenes (烯烃) vs. Alkynes (炔烃)

Alcohols (醇类): -OH functional group

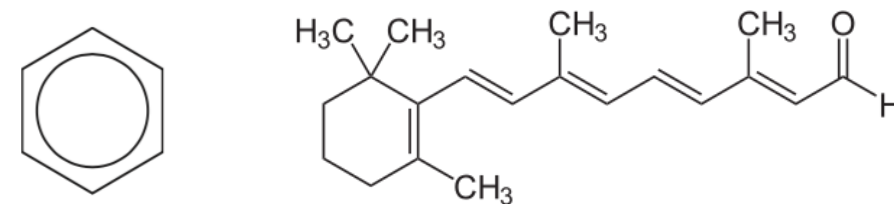
Phenols (苯酚类): -OH group is attached directly to an aromatic ring

Ethers (醚类): -O- functional group

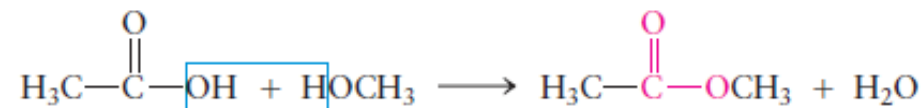
Aldehydes (醛类) vs. Ketones (酮类) vs. Carboxylic Acids (羧酸)



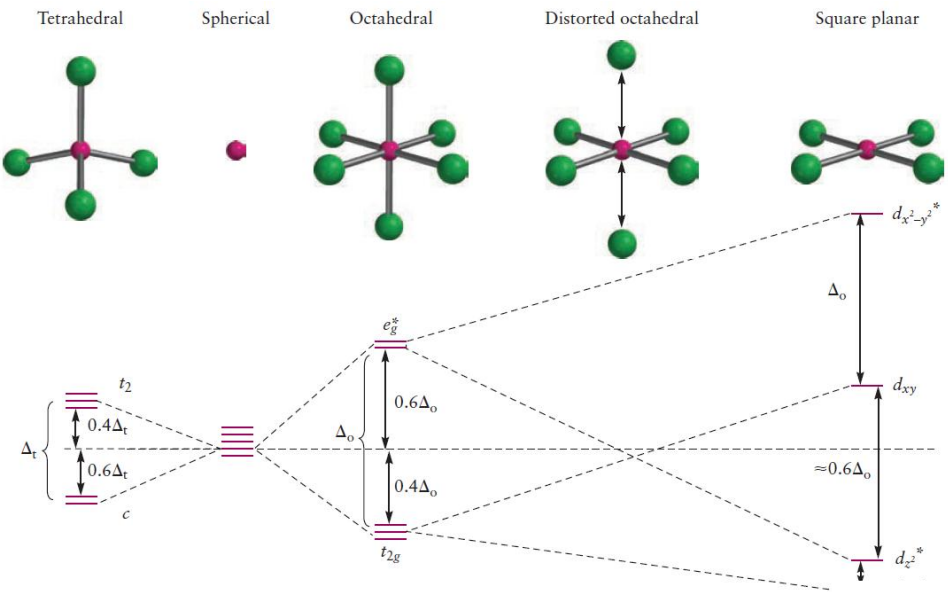
Conjugate system (共轭体系)



Esters 羧酸酯



8. Bonding in transition metal compounds and coordination complexes



Octahedral: (most common):

- the formation of 6 bonds, rather than 4, confers greater stability.

Tetrahedral: less stable

- The crystal field splitting is smaller
- The lower-energy set of levels is only doubly degenerate

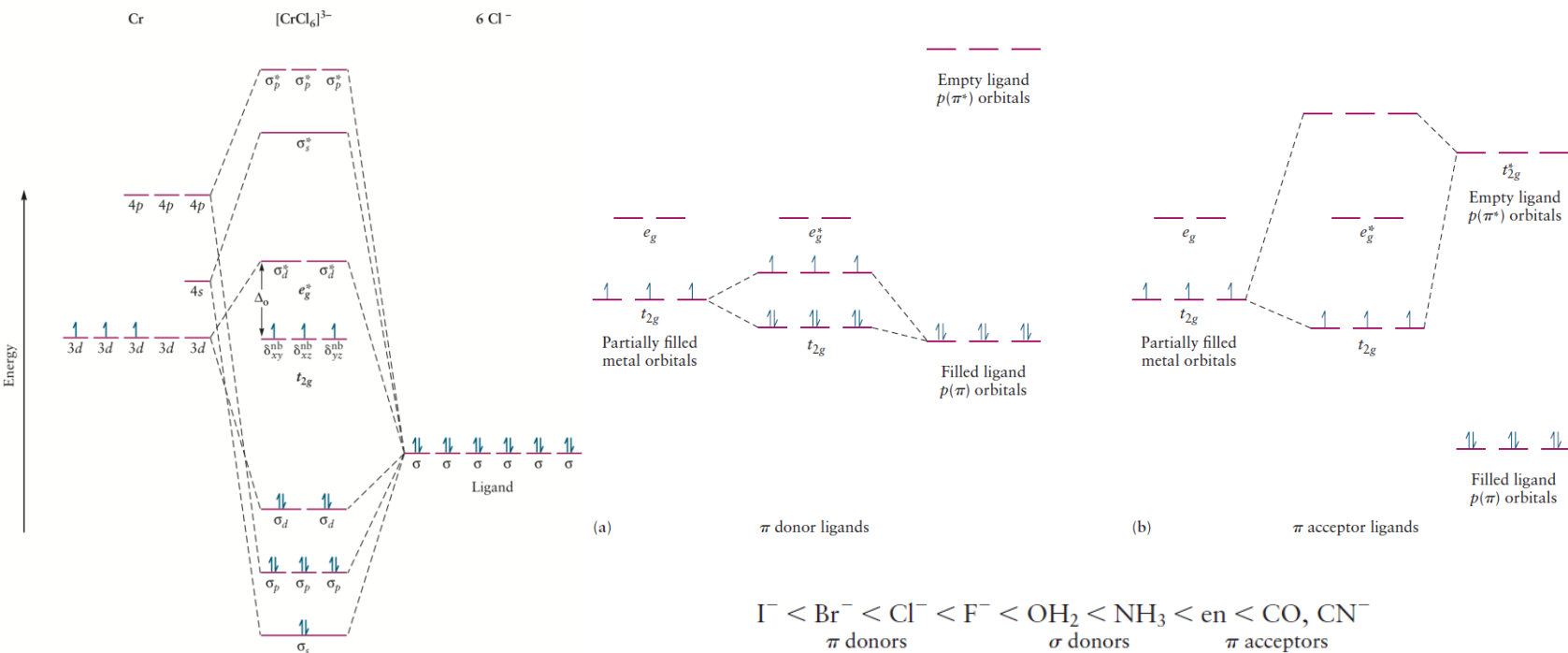
Square-planar: d⁸ ions & strong field ligands.

- The low-spin configuration leaves the high-energy d_{x²-y²} orbital vacant is the most stable.

TABLE 8.7
Examples of Hybrid Orbitals and Bonding in Complexes

Coordination Number	Hybrid Orbital	Configuration	Examples
2	sp	Linear	[Ag(NH ₃) ₂] ⁺
3	sp ²	Trigonal planar	BF ₃ , NO ₃ ⁻ , [Ag(PR ₃) ₃] ⁺
4	sp ³	Tetrahedral	Ni(CO) ₄ , [MnO ₄] ⁻ , [Zn(NH ₃) ₄] ²⁺
4	dsp ²	Square planar	[Ni(CN) ₄] ²⁻ , [Pt(NH ₃) ₄] ²⁺
5	dsp ³	Trigonal bipyramidal	TaF ₅ , [CuCl ₅] ³⁻ , [Ni(PEt ₃) ₂ Br ₃]
6	d ² sp ³	Octahedral	[Co(NH ₃) ₆] ³⁺ , [PtCl ₆] ²⁻

From G.E. Kimball, Directed valence. *J. Chem. Phys.* 1940, 8, 188.



$I^- < Br^- < Cl^- < F^- < OH_2 < NH_3 < en < CO, CN^-$
π donors σ donors π acceptors

9. Molecular spectroscopy and photochemistry

Born–Oppenheimer approximation: slow nuclei, fast electrons.

$$E_{\text{total}} = E_{\text{trans}} + E_{\text{rot}} + E_{\text{vib}} + E_{\text{el}}$$

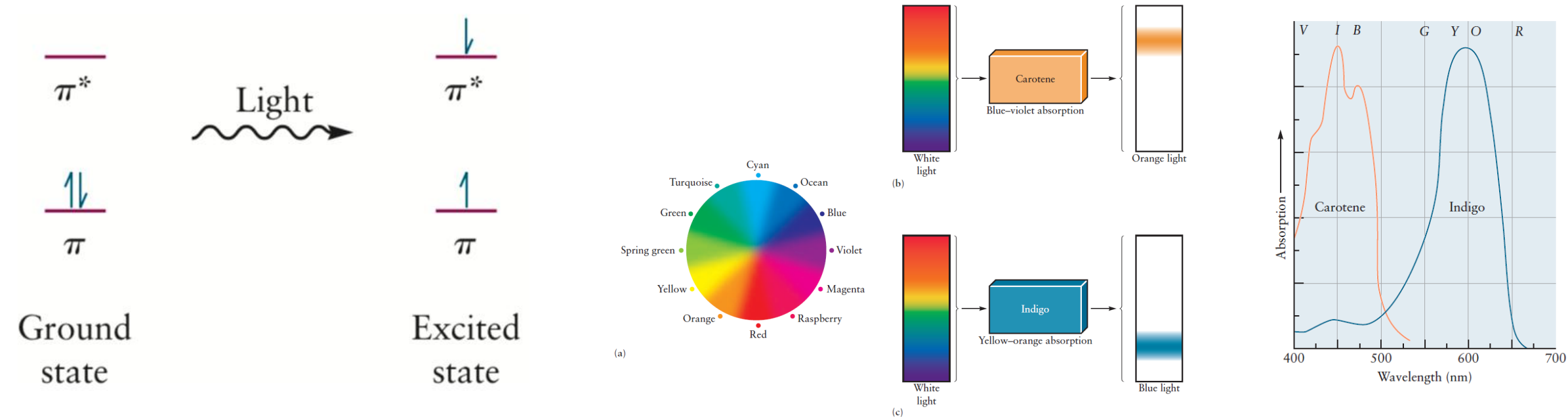
Spectroscopy (光谱): the interactions between electromagnetic radiation and matter.

Absorbed 吸收 Emitted 发射 Scattered 散射

$$\Delta E = h\nu_i$$

$$\Delta E = h\nu_i - h\nu_s$$

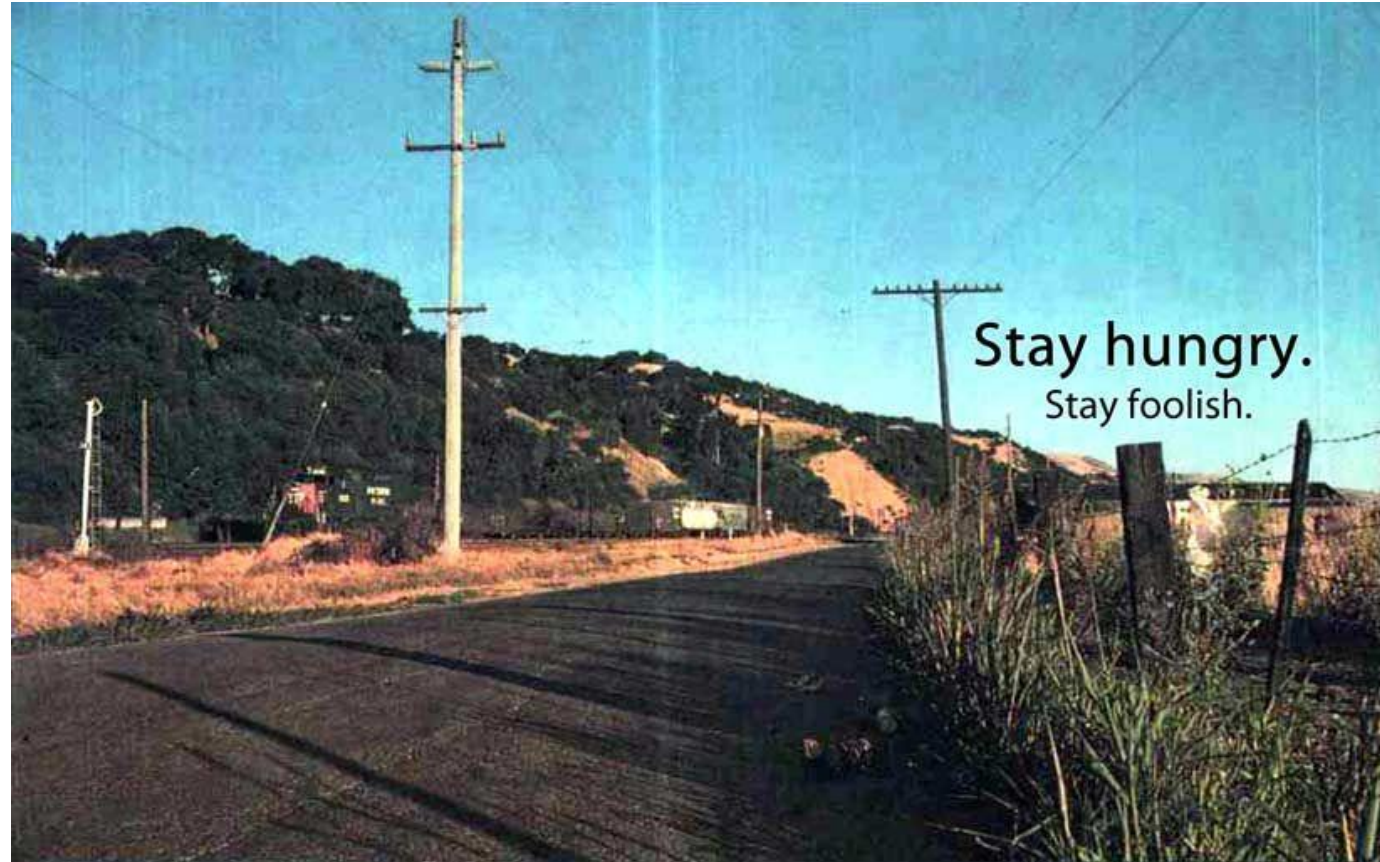
The intensity of the transmitted beam: **Beer–Lambert law**



Thank you for a great semester!

You can't connect the dots looking forward;
you can only connect them looking backward.
**So you have to trust that the dots will somehow
connect in your future.** You have to trust in
something — your gut, destiny, life, karma,
whatever. This approach has never let me down,
and it has made all the difference in my life.

Steve Jobs
2005, commencement address, Stanford University



“The Whole Earth Catalog”, 1972